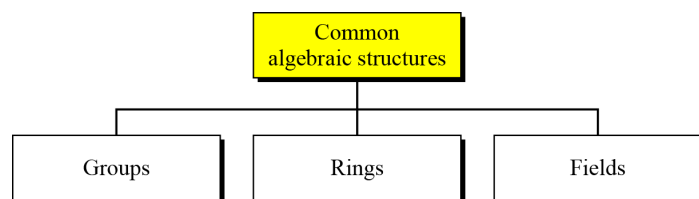


Algebraic Structures

- Cryptography requires sets of **integers** and **specific operations** that are defined for those **sets**.
- The combination of **the set and the operations** that are applied to the elements of the set is called an **algebraic structure**.
- In this section, we will define three common algebraic structures:

- **Groups**
- **Rings**
- **Fields**



Groups (G)

- A group (G) is a **set of elements with a binary operation ($\bullet$) that satisfies four properties** (or axioms).
 - **Closure** : if a, b are in G , then $a \bullet b$ in G
 - **Associativity** : if a, b, c are in G , then $(a \bullet b) \bullet c = a \bullet (b \bullet c)$
 - **Existence of identity**: For all a in G , there is an identity element (e), such that $e \bullet a = a \bullet e = a$
 - **Existence of inverse**: For each a in G , there is an element a' called inverse of a , such that $a \bullet a' = a' \bullet a = e$
- A commutative group (**Abelian Group**) satisfies an extra property, commutativity.
 - **Commutativity** : For all a and b in G , $a \bullet b = b \bullet a$ (Abelian G)

Groups

Properties

1. Closure
2. Associativity
3. Commutativity (See note)
4. Existence of identity
5. Existence of inverse

Note:

The third property needs to be satisfied only for a commutative group.

{a, b, c, ...}

Set



Operation

Operation:

1. Addition

2. Multiplication

Group

Groups: Application

- Although a group involves a single operation, the properties imposed on the operation allow the use of a **pair of operations as long as they are inverses of each other.**

Example:1

- The set of residue integers with the addition operator,

$$G = \langle \mathbb{Z}_n, + \rangle$$

- is a commutative group.
- We can perform addition and subtraction on the elements of this set without moving out of the set.

Groups: Application

Example:2

- The set Z_n^* with the multiplication operator, $G = \langle Z_n^*, \times \rangle$, is also an abelian group.
- Let us define a set $G = \langle \{a, b, c, d\}, \bullet \rangle$ and the operation as shown in Table below.

$\bullet$	a	b	c	d
a	a	b	c	d
b	b	c	d	a
c	c	d	a	b
d	d	a	b	c

This user defined operation $\bullet$ Forms an Abelian Group.

Groups: Finite Group

- A group is said to be a **finite group if the set has a finite number of elements**.
- Otherwise, it is an infinite group.
- Eg: $\langle \mathbb{Z}, + \rangle \rightarrow$ Infinite Group
- Eg: $\langle \mathbb{Z}_{10}, + \rangle \rightarrow$ Finite Group

Groups: Order of a Group

- The order of a group $|G|$, is **the number of elements in the group**.
- If the group is not finite, its order is infinite.
- If the group is finite, the order is finite.
- Eg: $G = \langle \mathbb{Z}_{10}, + \rangle$
 - $|G| = 10$

Groups: Sub Group

- A **subset H of a group G is a subgroup of G** if
 - **H itself is a group** with respect to the operation on G .
- If $G = \langle S, \bullet \rangle$ is a group, $H = \langle T, \bullet \rangle$ is a group under the same operation $\bullet$, and T is a non empty subset of S .
- This implies that, $H = \langle T, \bullet \rangle$ is a sub group of G .

Groups: Sub Group

- Is the group $H = \langle \mathbb{Z}_{10}, + \rangle$ a subgroup of the group $G = \langle \mathbb{Z}_{12}, + \rangle$?

Groups: Sub Group

- Is the group $H = \langle \mathbb{Z}_{10}, + \rangle$ a subgroup of the group $G = \langle \mathbb{Z}_{12}, + \rangle$?

Solution:

- The answer is no.
- Although H is a subset of G , the operations defined for these two groups are different.
- The operation in H is addition modulo 10; the operation in G is addition modulo 12.

Groups: Cyclic Subgroups

- If a subgroup of a **group can be generated using the power of an element**, the subgroup is called the **cyclic subgroup**.

$$a^n \rightarrow a \bullet a \bullet \dots \bullet a \quad (n \text{ times})$$

Groups: Cyclic Subgroups

- **Four cyclic subgroups can be made from the group $G = \langle \mathbb{Z}_6, + \rangle$.**
- They are $H_1 = \langle \{0\}, + \rangle$, $H_2 = \langle \{0, 2, 4\}, + \rangle$, $H_3 = \langle \{0, 3\}, + \rangle$, and $H_4 = G$.

$$0^0 \bmod 6 = 0$$

$$1^0 \bmod 6 = 0$$

$$1^1 \bmod 6 = 1$$

$$1^2 \bmod 6 = (1 + 1) \bmod 6 = 2$$

$$1^3 \bmod 6 = (1 + 1 + 1) \bmod 6 = 3$$

$$1^4 \bmod 6 = (1 + 1 + 1 + 1) \bmod 6 = 4$$

$$1^5 \bmod 6 = (1 + 1 + 1 + 1 + 1) \bmod 6 = 5$$

$$2^0 \bmod 6 = 0$$

$$2^1 \bmod 6 = 2$$

$$2^2 \bmod 6 = (2 + 2) \bmod 6 = 4$$

$$3^0 \bmod 6 = 0$$

$$3^1 \bmod 6 = 3$$

$$4^0 \bmod 6 = 0$$

$$4^1 \bmod 6 = 4$$

$$4^2 \bmod 6 = (4 + 4) \bmod 6 = 2$$

$$5^0 \bmod 6 = 0$$

$$5^1 \bmod 6 = 5$$

$$5^2 \bmod 6 = 4$$

$$5^3 \bmod 6 = 3$$

$$5^4 \bmod 6 = 2$$

$$5^5 \bmod 6 = 1$$

Groups: Cyclic Subgroups

- **Three cyclic subgroups** can be made from the group

$$G = \langle \mathbb{Z}_{10}^*, X \rangle.$$

- G has only four elements: 1, 3, 7, and 9.
- The cyclic subgroups are $H_1 = \langle \{1\}, \times \rangle$, $H_2 = \langle \{1, 9\}, \times \rangle$, and $H_3 = G$.

$$1^0 \bmod 10 = 1$$

$$3^0 \bmod 10 = 1$$

$$3^1 \bmod 10 = 3$$

$$3^2 \bmod 10 = 9$$

$$3^3 \bmod 10 = 7$$

$$7^0 \bmod 10 = 1$$

$$7^1 \bmod 10 = 7$$

$$7^2 \bmod 10 = 9$$

$$7^3 \bmod 10 = 3$$

$$9^0 \bmod 10 = 1$$

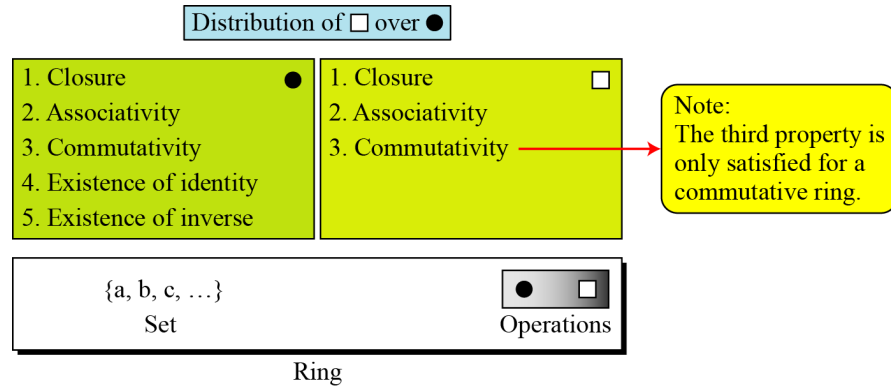
$$9^1 \bmod 10 = 9$$

Ring

- A **set of “numbers” with two operations** (addition and multiplication) which form:
- An **abelian group with addition operation** and
 - **Multiplication operation:**
 - has closure
 - is associative
 - **distributive over addition:** $a(b+c) = ab + ac$
- If multiplication operation is commutative, it forms a commutative ring

Ring

- A ring, $R = \langle \{...\}, \bullet, \square \rangle$, is an algebraic structure with two operations.

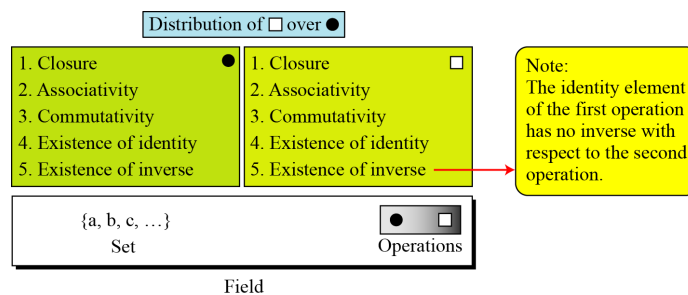


Ring

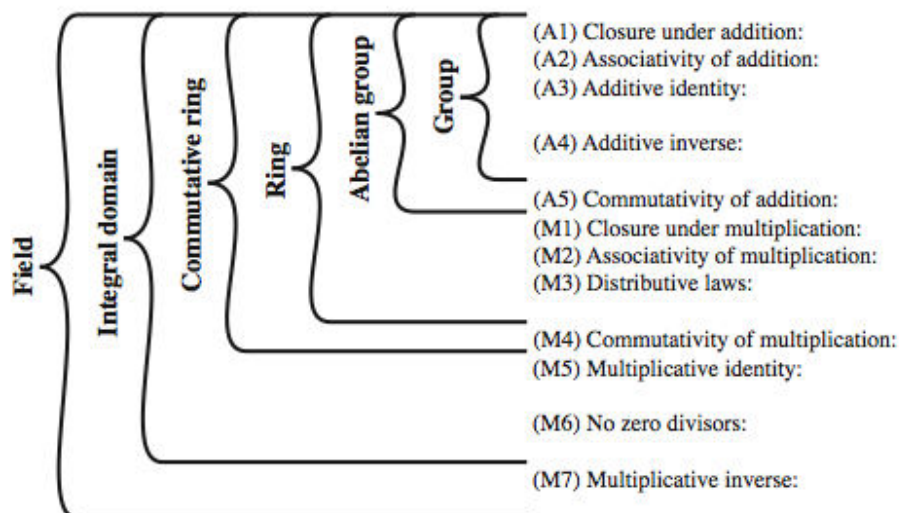
- **The set $\mathbb{Z}$ with two operations, addition and multiplication, is a commutative ring.**
- We show it by $R = \langle \mathbb{Z}, +, \times \rangle$.
- Addition satisfies all of the five properties
- Multiplication satisfies only three properties.

Field

- A set of numbers
- with two operations which form:
 - **abelian group for addition**
 - **abelian group for multiplication** (ignoring 0)
- A field, denoted by $F = \langle \{ \dots \}, \bullet, \square \rangle$ is a commutative ring in which the second operation satisfies all five properties defined for the first operation except that the identity of the first operation has no inverse.



Group, Ring, Field



Finite (Galois) Fields

- Galois showed that **for a field to be finite**, the **number of elements should be p^n** , where **p is a prime** and **n is a positive integer**.
- Denoted as **$GF(p^n)$**
- In particular often use the fields:
 - **$GF(p)$**
 - **$GF(2^n)$**

A Galois field, $GF(p^n)$, is a finite field with p^n elements.

Finite (Galois) Fields

- $GF(p)$ is the **set of integers $\{0, 1, \dots, p-1\}$** with arithmetic **operations modulo prime p** .
- These **form a finite field**
 - Since they have **multiplicative inverses**
 - Inverse can be found with Extended Euclidean algorithm
- Hence arithmetic is “well-behaved” and can do **addition, subtraction, multiplication, and division** without leaving the field $GF(p)$

Finite (Galois) Fields: GF(p) Fields

- When $n = 1$, we have GF(p) field.
- This field can be the set $\mathbb{Z}_p, \{0, 1, \dots, p - 1\}$, with two arithmetic operations.

Example:

- A very common field in this category is **GF(2)** with the set **{0, 1}** and two operations, addition and multiplication.

GF(2)

{0, 1}	+	×
--------	---	---

+	0	1
0	0	1
1	1	0

Addition

XOR

×	0	1
0	0	0
1	0	1

Multiplication

a	0	1
-a	1	0

a	0	1
a ⁻¹	—	1

Inverses

Finite (Galois) Fields: GF(p) Fields

Example:

- We can define GF(5) on the set $\mathbb{Z}_5$ (5 is a prime) with addition and multiplication operators

GF(5)

{0, 1, 2, 3, 4}	+	×
-----------------	---	---

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Addition

×	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Multiplication

Additive inverse

a	0	1	2	3	4
-a	0	4	3	2	1

a	0	1	2	3	4
a ⁻¹	—	1	3	2	4

Multiplicative inverse

Finite (Galois) Fields: GF(p) Fields

Summary

<i>Algebraic Structure</i>	<i>Supported Typical Operations</i>	<i>Supported Typical Sets of Integers</i>
Group	(+ -) or ($\times$ $\div$)	$\mathbf{Z}_n$ or $\mathbf{Z}_n^*$
Ring	(+ -) and ($\times$)	$\mathbf{Z}$
Field	(+ -) and ($\times$ $\div$)	$\mathbf{Z}_p$

GF(2^n) Fields

- In cryptography, we often need to use binary operations.
- In other words, we need to use fields.
- We can work in GF(2^n) and uses a set of 2^n elements.
- **The elements in this set are n-bit words.**

Topics discussed in this section:

- Polynomials (Polynomial Arithmetic)
- Using a Generator

GF(2ⁿ) Fields : Polynomial Arithmetic

- Let us define a GF(2²) field in which the set has four 2-bit words: {00, 01, 10, 11}.
- We can **redefine addition and multiplication for this field in such a way that all properties of these operations are satisfied.**

Addition

⊕	00	01	10	11
00	00	01	10	11
01	01	00	11	10
10	10	11	00	01
11	11	10	01	00

Identity: 00

Multiplication

⊗	00	01	10	11
00	00	00	00	00
01	00	01	10	11
10	00	10	11	01
11	00	11	01	10

Identity: 01

Based on Polynomial Multiplication

GF(2ⁿ) Fields : Polynomial Arithmetic

- A polynomial of degree $n - 1$ is an expression of the form

$$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x^1 + a_0x^0$$

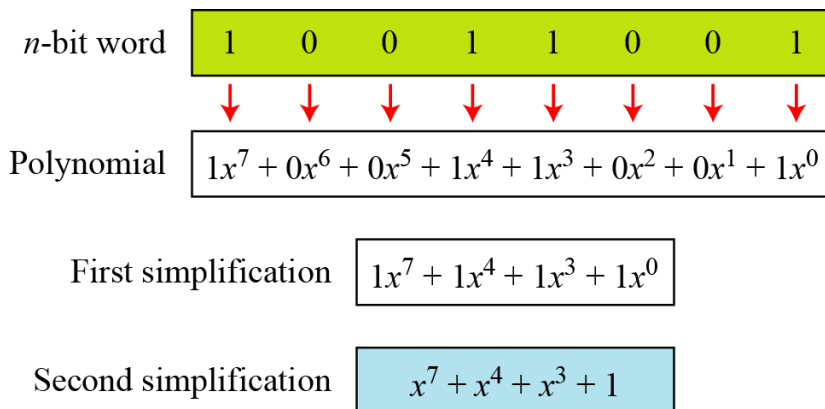
- where x^i is called the i^{th} term and a_i is called coefficient of the i^{th} term.

Several alternatives available:

- ordinary polynomial arithmetic
- poly arithmetic with coefs mod p
- poly arithmetic with coefs mod p and polynomials mod $m(x)$

GF(2ⁿ) Fields : Polynomial Arithmetic

- we can represent the 8-bit word (10011001) using a polynomials.



GF(2ⁿ) Fields : Polynomial Arithmetic

- To find the **8-bit word** related to the polynomial $x^5 + x^2 + x$, we first supply the omitted terms.
- Since $n = 8$, it means the polynomial is of degree 7.
- The expanded polynomial is

$$0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0$$

- This is related to the 8-bit word **00100110**.

Polynomial Arithmetic Operations

- It involves two operations:
 - Operation on Coefficients
 - Operation on two Polynomials
- i.e we have to **define two Fields, one for coefficients and one for the polynomial**
- **Since Coefficients are made of 0 or 1 we use GF(2) field**
- **For the Polynomials we use GF(2ⁿ) field**

Polynomials representing n -bit words use two fields: GF(2) and GF(2ⁿ).

Ordinary Polynomial Arithmetic

- Add or subtract corresponding coefficients
- multiply all terms by each other
- **Example:**
 - let $f(x) = x^3 + x^2 + 2$ and $g(x) = x^2 - x + 1$
 - $f(x) + g(x) = x^3 + 2x^2 - x + 3$
 - $f(x) - g(x) = x^3 + x + 1$
 - $f(x) \times g(x) = x^5 + 3x^2 - 2x + 2$

So, Ordinary polynomial arithmetic cannot be used for our computation.

As, Coefficients and Exponents are out of the defined limit !!!

Polynomial Arithmetic : Addition

- Let us do $(x^5 + x^2 + x) \oplus (x^3 + x^2 + 1)$ in $GF(2^8)$.
- We use the symbol $\oplus$ to show that we mean polynomial addition.
- The following shows the procedure:

$$\begin{array}{rcl}
 0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0 & \oplus & \\
 0x^7 + 0x^6 + 0x^5 + 0x^4 + 1x^3 + 1x^2 + 0x^1 + 1x^0 & & \\
 \hline
 0x^7 + 0x^6 + 1x^5 + 0x^4 + 1x^3 + 0x^2 + 1x^1 + 1x^0 & \rightarrow & x^5 + x^3 + x + 1
 \end{array}$$

XOR

Polynomial Arithmetic : Addition

- There is also another short cut. Because the addition in $GF(2)$ means the exclusive-or (XOR) operation.
- So we can exclusive-or the two words, bits by bits, to get the result.
- In the previous example,
 - $x^5 + x^2 + x$ is 00100110
 - $x^3 + x^2 + 1$ is 00001101.
- The result is 00101011 or
 - in polynomial notation $x^5 + x^3 + x + 1$.

Polynomial Arithmetic : Addition

- **Additive Identity:**

- 0 polynomial

- **Additive Inverse:**

- The polynomial itself

Addition and Subtraction Operations on polynomials are the same operations.

Polynomial Arithmetic : Multiplication

1. The coefficient multiplication is done in GF(2).
2. Multiplying x^i by x^j results in x^{i+j} .
3. The multiplication may create terms with degree more than $n - 1$, which means the **result needs to be reduced using a modulus polynomial**.

Polynomial Arithmetic : Modulus

- Addition of two polynomials doesn't create a polynomial out of a set. [XOR]
- But, in multiplication of two polynomials it may create a polynomial out of a set with degree more than $n-1$.
 - We need to divide the result by a modulus and keep only the remainder.
- For the sets of polynomials in $GF(2^n)$, a group of polynomials of degree n is defined as the modulus.
- The modulus in this case act as a prime polynomial
- Such polynomials are referred to as irreducible polynomials.

Polynomial Arithmetic : Modulus

List of irreducible polynomials:

Degree	Irreducible Polynomials
1	$(x + 1), (x)$
2	$(x^2 + x + 1)$
3	$(x^3 + x^2 + 1), (x^3 + x + 1)$
4	$(x^4 + x^3 + x^2 + x + 1), (x^4 + x^3 + 1), (x^4 + x + 1)$
5	$(x^5 + x^2 + 1), (x^5 + x^3 + x^2 + x + 1), (x^5 + x^4 + x^3 + x + 1), (x^5 + x^4 + x^3 + x^2 + 1), (x^5 + x^4 + x^2 + x + 1)$

Polynomial Arithmetic : Multiplication

- Find the result of $(x^5 + x^2 + x) \otimes (x^7 + x^4 + x^3 + x^2 + x)$ in $GF(2^8)$ with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$.
- Note that we use the symbol $\otimes$ to show the multiplication of two polynomials.

- Solution:**

$$\begin{aligned} P_1 \otimes P_2 &= x^5(x^7 + x^4 + x^3 + x^2 + x) + x^2(x^7 + x^4 + x^3 + x^2 + x) + x(x^7 + x^4 + x^3 + x^2 + x) \\ P_1 \otimes P_2 &= x^{12} + x^9 + x^8 + x^7 + x^6 + x^9 + x^6 + x^5 + x^4 + x^3 + x^8 + x^5 + x^4 + x^3 + x^2 \\ P_1 \otimes P_2 &= (x^{12} + x^7 + x^2) \bmod (x^8 + x^4 + x^3 + x + 1) = x^5 + x^3 + x^2 + x + 1 \end{aligned}$$

- To find the final result, divide the polynomial of degree 12 by the polynomial of degree 8 (the modulus) and keep only the remainder.

Polynomial Arithmetic : Multiplication

- Polynomial division with coefficients in $GF(2)$*

$$\begin{array}{r} x^4 + 1 \\ \hline x^8 + x^4 + x^3 + x + 1 \quad \left| \begin{array}{l} x^{12} + x^7 + x^2 \\ x^{12} + x^8 + x^7 + x^5 + x^4 \\ \hline x^8 + x^5 + x^4 + x^2 \\ x^8 + x^4 + x^3 + x + 1 \\ \hline \end{array} \right. \\ \hline \text{Remainder} \quad x^5 + x^3 + x^2 + x + 1 \end{array}$$

Polynomial Arithmetic : Multiplication

- **Multiplicative Identity:**

- $GF(2^8)=00000001$

- **Multiplicative Inverse:**

- Extended Euclidean Algorithm must be applied to the modulus and the polynomial.

Polynomial Arithmetic : Multiplication

Example:

- In $GF(2^4)$, find the inverse of $(x^2 + 1)$ modulo $(x^4 + x + 1)$.

Solution

- The answer is $(x^3 + x + 1)$ as shown in

Extended Euclidean algorithm

q	r_1	r_2	r	t_1	t_2	t
$(x^2 + 1)$	$(x^4 + x + 1)$	$(x^2 + 1)$	(x)	(0)	(1)	$(x^2 + 1)$
(x)	$(x^2 + 1)$	(x)	(1)	(1)	$(x^2 + 1)$	$(x^3 + x + 1)$
(x)	(x)	(1)	(0)	$(x^2 + 1)$	$(x^3 + x + 1)$	(0)
	(1)	(0)		$(x^3 + x + 1)$	(0)	

$$[(x^2 + 1) \otimes (x^3 + x + 1)] \bmod (x^4 + x + 1) = 1$$

Polynomial Arithmetic : Multiplication

- **Example:**

- In $GF(2^8)$, find the inverse of (x^5) modulo $(x^8 + x^4 + x^3 + x + 1)$.

Solution

- The answer is $(x^5 + x^4 + x^3 + x)$ as shown in

Extended Euclidean algorithm

q	r_1	r_2	r	t_1	t_2	t
(x^5)	$(x^8 + x^4 + x^3 + x + 1)$	(x^5)	$(x^4 + x^3 + x + 1)$	(0)	(1)	(x^3)
$(x + 1)$	(x^5)	$(x^4 + x^3 + x + 1)$	$(x^3 + x^2 + 1)$	(1)	(x^3)	$(x^4 + x^3 + 1)$
(x)	$(x^4 + x^3 + x + 1)$	$(x^3 + x^2 + 1)$	(1)	(x^3)	$(x^4 + x^3 + 1)$	$(x^5 + x^4 + x^3 + x)$
$(x^3 + x^2 + 1)$	$(x^3 + x^2 + 1)$	(1)	(0)	$(x^4 + x^3 + 1)$	$(x^5 + x^4 + x^3 + x)$	(0)
	(1)	(0)		$(x^5 + x^4 + x^3 + x)$	(0)	

$$[(x^5) \otimes (x^5 + x^4 + x^3 + x)] \bmod (x^8 + x^4 + x^3 + x + 1) = 1$$

Primes

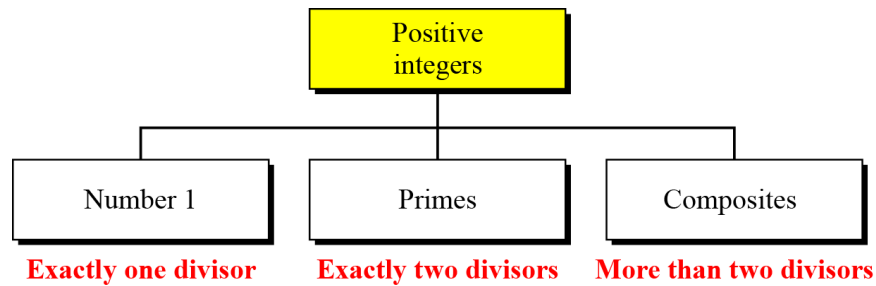
- **Asymmetric-key cryptography** uses primes extensively.

Topics discussed in this section:

- Definition
- Checking for Primes
- Fermat's Little Theorem
- Euler's Phi-Function
- Euler's Theorem

PRIMES: Definition

- Three groups of Positive Integers



A prime is divisible only by itself and 1.

PRIMES: Definition

- *What is the smallest prime?*

Solution

- The smallest prime is 2, which is divisible by 2 (itself) and 1.

- *List the primes smaller than 10.*

Solution

- There are four primes less than 10: 2, 3, 5, and 7.
- It is interesting to note that the percentage of primes in the range 1 to 10 is 40%.
- The percentage decreases as the range increases.

PRIMES: Checking for Primes

- Given a number n , how can we determine if n is a prime?
- The answer is that we need to see if the number is divisible by all primes less than

$$\sqrt{n}$$

- We know that this method is inefficient, but it is a good start.

PRIMES: Checking for Primes

Theorem

If n is composite, then n has a prime divisor less than or equal to $\sqrt{n}$.

Proof.

- Let $n = ab$, $1 < a < n$, $1 < b < n$.
- We can't have both $a > \sqrt{n}$ and $b > \sqrt{n}$ since this would lead to $ab > n$.
- Therefore, n must have a prime divisor less than or equal to $\sqrt{n}$.



PRIMES: Checking for Primes

- Is 97 a prime?

Solution

- The floor of $\sqrt{97} = 9$. The primes less than 9 are 2, 3, 5, and 7. We need to see if 97 is divisible by any of these numbers. It is not, so 97 is a prime.

- Is 301 a prime?

Solution

- The floor of $\sqrt{301} = 17$. We need to check 2, 3, 5, 7, 11, 13, and 17. The numbers 2, 3, and 5 do not divide 301, but 7 does. Therefore 301 is not a prime.

PRIMES: Checking for Primes

Table 9.1 *Sieve of Eratosthenes*

	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Fermat's (Little) Theorem

- **First Version**

$$a^{p-1} \equiv 1 \pmod{p}$$

- Multiply by a on both sides: **Second Version**

$$a^p \equiv a \pmod{p}$$

- where p is prime and $\gcd(a,p)=1$

Fermat's (Little) Theorem

- **Find the result of $6^{10} \pmod{11}$.**

Solution

$$a^{p-1} \equiv 1 \pmod{p}$$

- We have $6^{10} \pmod{11} = 1$.
- This is the first version of Fermat's little theorem where $p = 11$.

-
- **Find the result of $3^{12} \pmod{11}$.**

Solution

$$a^p \equiv a \pmod{p}$$

- Here the exponent (12) and the modulus (11) are not the same. With substitution this can be solved using Fermat's little theorem.

$$3^{12} \pmod{11} = (3^{11} \times 3) \pmod{11} = (3^{11} \pmod{11}) (3 \pmod{11}) = (3 \times 3) \pmod{11} = 9$$

Fermat's (Little) Theorem

- Multiplicative Inverses: [First version/a]

$$a^{-1} \bmod p = a^{p-2} \bmod p$$

- The answers to multiplicative inverses modulo a prime can be found without using the extended Euclidean algorithm:
 - a. $8^{-1} \bmod 17 = 8^{17-2} \bmod 17 = 8^{15} \bmod 17 = 15 \bmod 17$
 - b. $5^{-1} \bmod 23 = 5^{23-2} \bmod 23 = 5^{21} \bmod 23 = 14 \bmod 23$
 - c. $60^{-1} \bmod 101 = 60^{101-2} \bmod 101 = 60^{99} \bmod 101 = 32 \bmod 101$
 - d. $22^{-1} \bmod 211 = 22^{211-2} \bmod 211 = 22^{209} \bmod 211 = 48 \bmod 211$

Euler's Phi-Function

- Euler's phi-function, $\phi(n)$, which is sometimes called the Euler's **totient** function plays a very important role in cryptography.
 1. $\phi(1) = 0$.
 2. $\phi(p) = p - 1$ if p is a prime.
 3. $\phi(m \times n) = \phi(m) \times \phi(n)$ if m and n are relatively prime.
 4. $\phi(p^e) = p^e - p^{e-1}$ if p is a prime.

Euler's Phi-Function

- We can combine the above four rules to find the value of $\phi(n)$.
- For example, if n can be factored as

$$n = p_1^{e_1} \times p_2^{e_2} \times \dots \times p_k^{e_k}$$

- we combine the **third** and the **fourth** rule to find $\phi(n)$

$$\phi(n) = (p_1^{e_1} - p_1^{e_1-1}) \times (p_2^{e_2} - p_2^{e_2-1}) \times \dots \times (p_k^{e_k} - p_k^{e_k-1})$$

The difficulty of finding $\phi(n)$ depends on the difficulty of finding the factorization of n .

Euler's Phi-Function

- *What is the value of $\phi(13)$?*

Solution

- Because 13 is a prime, $\phi(13) = (13 - 1) = 12$.

- *What is the value of $\phi(10)$?*

Solution

- We can use the third rule: $\phi(10) = \phi(2) \times \phi(5) = 1 \times 4 = 4$, because 2 and 5 are primes.

1. $\phi(1) = 0$.
2. $\phi(p) = p - 1$ if p is a prime.
3. $\phi(m \times n) = \phi(m) \times \phi(n)$
4. $\phi(p^e) = p^e - p^{e-1}$ if p is a prime.

Euler's Phi-Function

- What is the value of $\phi(240)$?

Solution

- We can write $240 = 2^4 \times 3^1 \times 5^1$. Then
 - $$\phi(240) = (2^4 - 2^3) \times (3^1 - 3^0) \times (5^1 - 5^0) = 64$$
-

- Can we say that $\phi(49) = \phi(7) \times \phi(7) = 6 \times 6 = 36$?

Solution

- No. The third rule applies when m and n are relatively prime. Here $49 = 7^2$.
- We need to use the fourth rule: $\phi(49) = 7^2 - 7^1 = 42$.

1. $\phi(1) = 0$.
2. $\phi(p) = p - 1$ if p is a prime.
3. $\phi(m \times n) = \phi(m) \times \phi(n)$
4. $\phi(p^e) = p^e - p^{e-1}$ if p is a prime.

Euler's Phi-Function

- What is the number of elements in Z_{14}^* ?

Solution

- The answer is $\phi(14) = \phi(7) \times \phi(2) = 6 \times 1 = 6$. The members are 1, 3, 5, 9, 11, and 13.

Interesting point: If $n > 2$, the value of $\phi(n)$ is even.

Euler's Theorem

- First Version

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

- Second Version

$$a^{k \times \phi(n) + 1} \equiv a \pmod{n}$$

The second version of Euler's theorem is used in the RSA cryptosystem

Euler's Theorem

- Find the result of $6^{24} \pmod{35}$.

Solution

- We have $6^{24} \pmod{35} = 6^{\phi(35)} \pmod{35} = 1$.

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

- Find the result of $20^{62} \pmod{77}$.

Solution

- If we let $k = 1$ on the second version, we have
- $20^{62} \pmod{77} = (20 \pmod{77}) (20^{\phi(77) + 1} \pmod{77}) \pmod{77}$
- $= (20)(20) \pmod{77} = 15$.

$$a^{k \times \phi(n) + 1} \equiv a \pmod{n}$$

Euler's Theorem

Multiplicative Inverses:

- Euler's theorem can be used to find multiplicative inverses modulo a composite.

$$a^{-1} \bmod n = a^{\phi(n)-1} \bmod n$$

Euler's Theorem

$$a^{-1} \bmod n = a^{\phi(n)-1} \bmod n$$

- The answers to multiplicative inverses modulo a composite can be found without using the extended Euclidean algorithm if we know the factorization of the composite:

a. $8^{-1} \bmod 77 = 8^{\phi(77)-1} \bmod 77 = 8^{59} \bmod 77 = 29 \bmod 77$

b. $7^{-1} \bmod 15 = 7^{\phi(15)-1} \bmod 15 = 7^7 \bmod 15 = 13 \bmod 15$

c. $60^{-1} \bmod 187 = 60^{\phi(187)-1} \bmod 187 = 60^{159} \bmod 187 = 53 \bmod 187$

d. $71^{-1} \bmod 100 = 71^{\phi(100)-1} \bmod 100 = 71^{39} \bmod 100 = 31 \bmod 100$
 $(2^2) \cdot (5^2)$

Chinese Remainder Theorem

The Chinese remainder theorem (CRT) is used to solve a set of **congruent equations with one variable but different moduli**, which are relatively prime, as shown below:

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2} \\&\dots \\x &\equiv a_k \pmod{m_k}\end{aligned}$$

Chinese Remainder Theorem

- It is used to **speed up modulo computations**
 - if modulo can be represented as a product of numbers
 - eg. $\text{mod } M = m_1 m_2 \dots m_k$
- Chinese Remainder theorem lets us work in each **moduli m_i** separately
- **Since computational cost is proportional to size, this is faster than working in the full modulus M**

Chinese Remainder Theorem

Example:

- The following is an example of a set of equations with different moduli:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

- Find x ? (Trial and Error Method)

Chinese Remainder Theorem

Example:

- The following is an example of a set of equations with different moduli:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

- Find x ? (Trial and Error Method)
- The answer to this set of equations is $x = 23$.
- This value satisfies all equations: $23 \equiv 2 \pmod{3}$, $23 \equiv 3 \pmod{5}$, and $23 \equiv 2 \pmod{7}$.

Chinese Remainder Theorem

Solution using Chinese Remainder Theorem: (steps)

1. Find $M = m_1 \times m_2 \times \dots \times m_k$. This is the common modulus.
2. Find $M_1 = M/m_1, M_2 = M/m_2, \dots, M_k = M/m_k$.
3. Find the multiplicative inverse of $M_1, M_2, \dots, M_k$ using the corresponding moduli $(m_1, m_2, \dots, m_k)$. Call the inverses $M_1^{-1}, M_2^{-1}, \dots, M_k^{-1}$.
4. The solution to the simultaneous equations is

$$x = (a_1 \times M_1 \times M_1^{-1} + a_2 \times M_2 \times M_2^{-1} + \dots + a_k \times M_k \times M_k^{-1}) \bmod M$$

Chinese Remainder Theorem

$$\begin{aligned}x &\equiv 2 \pmod{3} \\x &\equiv 3 \pmod{5} \\x &\equiv 2 \pmod{7}\end{aligned}$$

Solution: Follow the four steps

1. $M = 3 \times 5 \times 7 = 105$
2. $M_1 = 105 / 3 = 35, M_2 = 105 / 5 = 21, M_3 = 105 / 7 = 15$
3. The inverses are $M_1^{-1} = 2, M_2^{-1} = 1, M_3^{-1} = 1$
 1. $M_1^{-1} \bmod m_1 ; M_2^{-1} \bmod m_2 ; M_3^{-1} \bmod m_3$
4. $x = (2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1) \bmod 105 = 23 \bmod 105$

Chinese Remainder Theorem

Example:

- Find an integer that has a remainder of 3 when divided by 7 and 13, but is divisible by 12.

Chinese Remainder Theorem

Example:

- Find an integer that has a remainder of 3 when divided by 7 and 13, but is divisible by 12.

Solution:

- This is a CRT problem. We can form three equations and solve them to find the value of x.

$$\begin{aligned}x &= 3 \bmod 7 \\x &= 3 \bmod 13 \\x &= 0 \bmod 12\end{aligned}$$

- we find $x = 276$.
- We can check that $276 = 3 \bmod 7$, $276 = 3 \bmod 13$ and 276 is divisible by 12 (the quotient is 23 and the remainder is zero).

Chinese Remainder Theorem

Example:

- Assume we need to calculate $z = x + y$ where $x = 123$ and $y = 334$, but our system accepts only numbers less than 100.

Solution:

$x \equiv 24 \pmod{99}$	$y \equiv 37 \pmod{99}$
$x \equiv 25 \pmod{98}$	$y \equiv 40 \pmod{98}$
$x \equiv 26 \pmod{97}$	$y \equiv 43 \pmod{97}$

- Adding each congruence in x with the corresponding congruence in y gives

$x + y \equiv 61 \pmod{99}$	$\rightarrow z \equiv 61 \pmod{99}$
$x + y \equiv 65 \pmod{98}$	$\rightarrow z \equiv 65 \pmod{98}$
$x + y \equiv 69 \pmod{97}$	$\rightarrow z \equiv 69 \pmod{97}$

- $z = 457$.

Primality Testing

- In cryptography **large prime numbers** are used in the process of encryption and decryption.
- When **we fail to generate larger prime numbers**, we can **randomly generate a large number and check if it is prime or not**.
 - Primality testing comes handy for this.
- Finding an algorithm to correctly and efficiently test a very large integer and find if its prime or composite is a challenging task.
- The algorithms that are developed for this issue is categorized into two,
 - **Deterministic Algorithm** – It always gives the correct answer
 - **Probabilistic Algorithm** – It gives the answer that is correct most of the times

1.Deterministic Algorithm

- It accepts an integer and outputs if it is prime or composite.
- It was found that, **all deterministic algorithms are inefficient at finding larger prime numbers.**
- So, the deterministic algorithms are considered **infeasible**.

1.1 Divisibility Algorithm

- Most **elementary technique to** find if the number is prime or not.
- We divide the given number with **all divisors less** than **square root of n**.
- **If all the numbers does not divide it, it is prime.**
- If any one divides it, it is a composite number.

-
- This was later improved as follows,
 - We divide the given number with **all prime divisors less** than **square root of n**.
 - **If all the numbers does not divide it, it is prime.**
 - If any one divides it, it is a composite number.

2. Probabilistic Algorithms

- Algorithms to test the primality can either be **decisional or randomized**.
- Yes or no answer – Decisional Problems
- Make random choices – Randomized Algorithms
 - **Monte Carlo Algorithms**
 - It is a Yes or No biased Algorithm
 - **Las Vegas Algorithm**
 - Algorithm may not give any answer, but if it gives an answer it will be correct.
- **A probabilistic algorithm does not guarantee the correctness of the result.**
 - **But we can make the probability of error small.**

2.1 Fermat Test

If n is a prime number, the following holds,

$$a^{n-1} \equiv 1 \pmod{n}$$

Where, $1 \leq a \leq n-1$

False positive results are possible when we apply fermat test.

If the fermat test concludes a composite number as a prime number, we refer to this number as **Carmichael Number**.

So, the fermat test only **returns a pseudo prime number or a composite number**.

Fermat Test

- Is 5 prime?

1 ≤ a ≤ n-1 → a takes the values [1,2,3,4]		
a=1	$1^{(5-1)} \bmod 5 = 1^4 \bmod 5 = 1$	Pass
a=2	$2^{(5-1)} \bmod 5 = 2^4 \bmod 5 = 16 \bmod 5 = 1$	Pass
a=3	$3^{(5-1)} \bmod 5 = 3^4 \bmod 5 = 81 \bmod 5 = 1$	Pass
a=4	$4^{(5-1)} \bmod 5 = 4^4 \bmod 5 = 256 \bmod 5 = 1$	Pass

5 is a prime number !!!

Fermat Test

- Is 7 prime?

1 ≤ a ≤ n-1 → a takes the values [1,2,3,4,5,6]		
a=1	$1^{(7-1)} \bmod 7 = 1^6 \bmod 7 = 1$	Pass
a=2	$2^{(7-1)} \bmod 7 = 2^6 \bmod 7 = 64 \bmod 7 = 1$	Pass
a=3	$3^{(7-1)} \bmod 7 = 3^6 \bmod 7 = 729 \bmod 7 = 1$	Pass
a=4	$4^{(7-1)} \bmod 7 = 4^6 \bmod 7 = 4096 \bmod 7 = 1$	Pass
a=5	$5^{(7-1)} \bmod 7 = 5^6 \bmod 7 = 15625 \bmod 7 = 1$	Pass
a=6	$6^{(7-1)} \bmod 7 = 6^6 \bmod 7 = 46656 \bmod 7 = 1$	Pass

7 is a prime number !!!

Fermat Test

When the numbers are big, the values of a are chosen randomly and checked.

Algorithm 1: Fermat Primality Test

Data: n : number to be tested for primality, k : number of times the test will be repeated for

Result: Composite or Pseudoprime

Function $isPrime(n, k)$ **is**

repeat k **times**

$a \leftarrow$ uniform random number between 2 and $n-1$;

if $a^{n-1} \not\equiv 1 \pmod{n}$ **then**

 return Composite;

end

end

 return Pseudoprime;

end

561 – Carmichael Number

2.2 Rabin Miller Method

- Michael Oser Rabin, Garu Lee Miller
- Probabilistic Primality Test
 - Is the number probably prime or not.
- Similar to fermats test

Rabin Miller Method

Algorithm 9.2 Pseudocode for Miller-Rabin test

```

Miller_Rabin_Test ( $n, a$ )           //  $n$  is the number;  $a$  is the base.
{
    Find  $m$  and  $k$  such that  $n - 1 = m \times 2^k$ 
     $T \leftarrow a^m \bmod n$ 
    if ( $T = \pm 1$ ) return "a prime"
    for ( $i \leftarrow 1$  to  $k - 1$ )           //  $k - 1$  is the maximum number of steps.
    {
         $T \leftarrow T^2 \bmod n$ 
        if ( $T = +1$ ) return "a composite"
        if ( $T = -1$ ) return "a prime"
    }
    return "a composite"
}

```

Rabin Miller Method – Example 1

n= 561		
Find the values of k and m	$n-1 = 2^k \cdot m$	$561-1=560$ $= 2^4 \cdot 35$ $k=4 \ m=35$
Decide on the value of a	$1 < a < n-1$	a= 2
Initial Computation	$T = a^m \bmod n$ T is not equal to +1 or -1	$T = 2^{35} \bmod 561$ $= 263$
Loop Count: 1 to k-1	1 to 3	
1	$T = T^2 \bmod n$	$T = 263^2 \bmod 561$ $= 166$
2	$T = T^2 \bmod n$	$T = 166^2 \bmod 561$ $= 67$
3	$T = T^2 \bmod n$	$T = 67^2 \bmod 561$ $= 1$
561 is a composite Number !!!		

Rabin Miller Method – Example 2

n= 61		
Find the values of k and m	$n-1 = 2^k \cdot m$	$61-1= 60$ $= 2^2 \cdot 15$ $k=2 \ m=15$
Decide on the value of a	$1 < a < n-1$	a= 2
Initial Computation	$T = a^m \bmod n$ T is not equal to +1 or -1	$T = 2^{15} \bmod 61$ $= 11$
Loop Count: 1 to k-1	1 to 1	
1	$T = T^2 \bmod n$	$T = 11^2 \bmod 61$ $= 60 \bmod 61 = -1$
61 is a prime Number !!!		

Rabin Miller Method – Example 3

n= 27		
Find the values of k and m	$n-1 = 2^k \cdot m$	$27-1= 26$ $= 2^1 \cdot 13$ $k=1 \ m=13$
Decide on the value of a	$1 < a < n-1$	a= 2
Initial Computation	$T = a^m \bmod n$ T is not equal to +1 or -1	$T = 2^{13} \bmod 27$ $= 11$
Loop Count: 1 to k-1	1 to 0 → No looping	
27 is a Composite Number !!!		

Recommended Primality Test

- Today, one of the most popular primality test is a combination of the divisibility test and the Miller-Rabin test.
- Following are the recommended steps:
 1. Choose an odd integer, because all even integers (except 2) are definitely composites.
 2. Do some **trivial divisibility tests on some known primes such as 3, 5, 7, 11, 13**, and so on to be sure that you are not dealing with an obvious composite. If the number passes all of these tests, move to the next step. If the number fails any of these tests, go back to step 1 and choose another odd number.
 3. **Choose a set of bases for testing.** A large set of bases is preferable.
 4. **Do Miller-Rabin tests on each of the bases.** If any of them fails, go back to step 1 and choose another odd number. If the **test passes for all bases, declare the number a strong pseudoprime.**

Recommended Primality Test

- The number 4033 is a composite (37×109). Does it pass the recommended primality test?

Solution

1. Perform the divisibility tests first. The numbers 2, 3, 5, 7, 11, 17, and 23 are not divisors of 4033.
2. Perform the Miller-Rabin test with a base of 2, $4033 - 1 = 63 \times 2^6$, which means m is 63 and k is 6.

Initialization: $T \equiv 2^{63} \pmod{4033} \equiv 3521 \pmod{4033}$
 $k = 1$ $T \equiv T^2 \equiv 3521^2 \pmod{4033} \equiv -1 \pmod{4033} \rightarrow \text{Passes}$

3. But we are not satisfied. We continue with another base, 3.

Recommended Primality Test

Initialization: $T \equiv 3^{63} \pmod{4033} \equiv 3551 \pmod{4033}$

$k = 1$	$T \equiv T^2 \equiv 3551^2 \pmod{4033} \equiv 2443 \pmod{4033}$
$k = 2$	$T \equiv T^2 \equiv 2443^2 \pmod{4033} \equiv 3442 \pmod{4033}$
$k = 3$	$T \equiv T^2 \equiv 3442^2 \pmod{4033} \equiv 2443 \pmod{4033}$
$k = 4$	$T \equiv T^2 \equiv 2443^2 \pmod{4033} \equiv 3442 \pmod{4033}$
$k = 5$	$T \equiv T^2 \equiv 3442^2 \pmod{4033} \equiv 2443 \pmod{4033} \rightarrow \text{Failed (composite)}$

Discrete Logarithm

- In cryptography, we also need to discuss modular logarithm.
- If exponentiation is used for encryption or decryption, the attacker can use logarithm to attack.
- We need to know how hard it is to reverse the exponentiation.
- Method: **Discrete Logarithm**
- To understand the concept of Discrete Logarithm we should understand certain properties of multiplicative groups.
 - Finite Multiplicative Group
 - Order of the Group
 - Order of an Element

To solve : $x = \log_a y \pmod{n}$
Solution is : $y = a^x \pmod{n}$

Discrete Logarithm

Finite Multiplicative Group:

- $G = \langle \mathbb{Z}_n^*, X \rangle$, is the Finite Multiplicative Group.
- $\mathbb{Z}_n^*$: has integers from 1 to $n-1$ that are relatively prime to n
- If the modulus of the Group is prime, then $G = \langle \mathbb{Z}_p^*, X \rangle$

Discrete Logarithm

Order of the Group:

- Order of a Finite Group is, $|G|$
- $|G|$, to be the number of elements in the Group G
- $G = \langle \mathbb{Z}_n^*, X \rangle$, the order of the group will be $\phi(n)$

Example:

- What is the order of group $G = \langle \mathbb{Z}_{21}^*, X \rangle$? $|G| = \phi(21) = \phi(3) \times \phi(7) = 2 \times 6 = 12$.
- There are 12 elements in this group: 1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, and 20.
- All are relatively prime with 21.

Discrete Logarithm

Order of an Element:

- Order of an Element a is, $\text{ord}(a)$
- $\text{ord}(a)$, is the smallest integer 'i' such that $a^i \equiv e \pmod{n}$
 - where, e is the identity element

Example:

- Find the order of all elements in $G = \langle \mathbb{Z}_{10}^*, \times \rangle$.
- This group has only $\phi(10) = 4$ elements: 1, 3, 7, 9.
- We can find the order of each element by trial and error.
- $1^1 \equiv 1 \pmod{10} \rightarrow \text{ord}(1) = 1$.
- $3^4 \equiv 1 \pmod{10} \rightarrow \text{ord}(3) = 4$.
- $7^4 \equiv 1 \pmod{10} \rightarrow \text{ord}(7) = 4$.
- $9^2 \equiv 1 \pmod{10} \rightarrow \text{ord}(9) = 2$.

Discrete Logarithm

Euler's Theorem:

- If 'a' is the member of $G = \langle \mathbb{Z}_n^*, \times \rangle$, then $a^{\phi(n)} \equiv 1 \pmod{n}$
- The relationship $a^i \equiv e \pmod{n}$ holds when $i = \phi(n)$.

Example:

- The result of $a^i \equiv x \pmod{8}$ for the group $G = \langle \mathbb{Z}_8^*, \times \rangle$
- $\phi(8) = 4$, i.e., (1,3,5,7)

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$
$a = 1$	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1
$a = 3$	x: 3	x: 1	x: 3	x: 1	x: 3	x: 1	x: 3
$a = 5$	x: 5	x: 1	x: 5	x: 1	x: 5	x: 1	x: 5
$a = 7$	x: 7	x: 1	x: 7	x: 1	x: 7	x: 1	x: 7

Discrete Logarithm

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$
$a = 1$	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1
$a = 3$	x: 3	x: 1	x: 3	x: 1	x: 3	x: 1	x: 3
$a = 5$	x: 5	x: 1	x: 5	x: 1	x: 5	x: 1	x: 5
$a = 7$	x: 7	x: 1	x: 7	x: 1	x: 7	x: 1	x: 7

The table reveals two points:

- The shaded area: when $i = \phi(8)$, the result is $x=1$ for every 'a'.
- The value of x can be 1 for many values of i .
- The first time when $x = 1$, the value of i gives the order of the element.

Discrete Logarithm

Primitive Roots

- In the group $G = \langle \mathbb{Z}_n^*, \times \rangle$, when the order of an element is the same as $\phi(n)$, that **element** is called the **primitive root of the group**.
- Example:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$
$a = 1$	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1
$a = 3$	x: 3	x: 1	x: 3	x: 1	x: 3	x: 1	x: 3
$a = 5$	x: 5	x: 1	x: 5	x: 1	x: 5	x: 1	x: 5
$a = 7$	x: 7	x: 1	x: 7	x: 1	x: 7	x: 1	x: 7

- The table shows that there are no primitive roots in $G = \langle \mathbb{Z}_8^*, \times \rangle$ because no element has the order equal to $\phi(8) = 4$.
- The order of elements are all smaller than 4.

Discrete Logarithm

Primitive Roots

- The result of $a^i \equiv x \pmod{7}$ for the group $G = \langle \mathbb{Z}_7^*, \times \rangle$ is shown below. In this group, $\phi(7) = 6$.

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$
$a = 1$	x: 1	x: 1	x: 1	x: 1	x: 1	x: 1
$a = 2$	x: 2	x: 4	x: 1	x: 2	x: 4	x: 1
Primitive root → $a = 3$	x: 3	x: 2	x: 6	x: 4	x: 5	x: 1
$a = 4$	x: 4	x: 2	x: 1	x: 4	x: 2	x: 1
Primitive root → $a = 5$	x: 5	x: 4	x: 6	x: 2	x: 3	x: 1
$a = 6$	x: 6	x: 1	x: 6	x: 1	x: 6	x: 1

Discrete Logarithm

- Primitive Roots

The group $G = \langle \mathbb{Z}_n^*, \times \rangle$ has primitive roots only if n is 2, 4, p^t , or $2p^t$.

Example

- For which value of n , does the group $G = \langle \mathbb{Z}_n^*, \times \rangle$ have primitive roots: 17, 20, 38, and 50?

Solution

- $G = \langle \mathbb{Z}_{17}^*, \times \rangle$ has primitive roots, 17 is a prime.
- $G = \langle \mathbb{Z}_{20}^*, \times \rangle$ has no primitive roots.
- $G = \langle \mathbb{Z}_{38}^*, \times \rangle$ has primitive roots, $38 = 2 \times 19$ prime.
- $G = \langle \mathbb{Z}_{50}^*, \times \rangle$ has primitive roots, $50 = 2 \times 5^2$ and 5 is a prime.

Discrete Logarithm

If the group $G = \langle \mathbb{Z}_n^*, \times \rangle$ has any primitive root, the number of primitive roots is $\phi(\phi(n))$.

Discrete Logarithm

Cyclic Group

- If g is a primitive root in the group, we can generate the set $\mathbb{Z}_n^*$ as $\mathbb{Z}_n^* = \{g^1, g^2, g^3, \dots, g^{\phi(n)}\}$.

Example:

- The group $G = \langle \mathbb{Z}_{10}^*, \times \rangle$ has two primitive roots because $\phi(10) = 4$ and $\phi(\phi(10)) = 2$.
- It can be found that the primitive roots are 3 and 7.
- The following shows how we can create the whole set $\mathbb{Z}_{10}^*$ using each primitive root.

$g = 3 \rightarrow$	$g^1 \bmod 10 = 3$	$g^2 \bmod 10 = 9$	$g^3 \bmod 10 = 7$	$g^4 \bmod 10 = 1$
$g = 7 \rightarrow$	$g^1 \bmod 10 = 7$	$g^2 \bmod 10 = 9$	$g^3 \bmod 10 = 3$	$g^4 \bmod 10 = 1$

Discrete Logarithm

The group $G = \langle \mathbb{Z}_n^*, \times \rangle$ is a cyclic group if it has primitive roots.
The group $G = \langle \mathbb{Z}_p^*, \times \rangle$ is always cyclic.

Discrete Logarithm

The idea of Discrete Logarithm

• Properties of $G = \langle \mathbb{Z}_p^*, \times \rangle$:

1. Its elements include all integers from 1 to $p - 1$.
2. It always has primitive roots.
3. It is cyclic. The elements can be created using g^x where x is an integer from 1 to $\phi(n) = p - 1$.
4. The primitive roots can be thought as the base of logarithm.